MIL-M-46314B(MU)
13 January 1969
SUPERSEDING
MIL-M-46314A(MU)
16 October 1963

MILITARY SPECIFICATION

MOUNT, TELESCOPE: M145

1. SCOPE

1.1 This specification covers one basic type of telescope mount primarily designed to provide adjustments in elevation and to compensate for angles of cant for indirect fire.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-F-13926	Fire Control Materiel: General Specification Governing the Manufacture and Inspection of
MIL-P-14232	Parts Equipment and Tools for Army Materiel; Packaging and Packing of
MIL-I-45607	Inspection Equipment, Supply and Maintenance of

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	Quality Assurance Terms and Definitions
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts

DRAWINGS

U.S. Army Frankford Arsenal

F8267701	Mount, Telescope: M145

PACKAGING DATA SHEET

8267701	Packaging of Mount, Telescope: M145

FSC 1240

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(Copies of specifications, standards, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Fabrication. - The mount shall be manufactured in accordance with Drawing F8267701 and drawings pertaining thereto. The requirements of this specification are detailed only to the extent considered necessary to insure inherent quality and performance capabilities.

3.2 General specification. - The contractor shall be responsible for adherence to, and compliance with, the following requirements of Specification MIL-F-13926:

(a) Order of precedence.
(b) Dimensions and tolerances.
(c) Inorganic protective surface finishes.
(d) Electrical and electronic assemblies.
(e) Part identification and marking.
(f) Workmanship.

3.3 Environmental.

3.3.1 Operating temperatures.- The mount shall meet the applicable torque requirements of 3.9 while exposed to and thermally stabilized at ambient temperatures of -40° and +150°F.

3.3.2 Vibration. - With the horizontal shaft immobilized, the mount shall be vibrated at a frequency of 30 cycles per second and an amplitude of 1/16 inch (1/8 inch total excursion) for a period of 5 minutes A 15 seconds. Subsequent to vibration, there shall be no evidence of physical failure and the mount shall meet the following requirements.

3.4 Orientation. - The requirements of 3.5 to 3.9 inclusive shall be met with the mount orientated in the following position:

(a) The flat mounting surface in a vertical plane.
(b) The axis of the horizontal shaft in a horizontal plane.
(c) The elevation and correction counters set at zero.
(d) The quadrant seat horizontal and its cross level vial centered.
(e) The telescope socket vertical within 0.5 mil as determined by the two keyways.
(f) Utilizing their respective mirrors for observation, the pitch and cant level vial bubbles shall be centered within the width. of a graduation line.
(g) Utilizing the mirror for observation, the elevation level vial bubble shall be centered within the width of a graduation 1ine.

NOTE: All level vials shall be illuminated for a minimum of 5 minutes prior to reading level vials.

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3.5 Performance.

3.5.1 Excursion ranges.

3.5.1.1 Cant. - Actuation of the cant correction knob shall move the telescope socket a minimum of 10 degrees in each direction from the position established in 3.4(e).

3.5.1.2 Pitch. - Actuation of the pitch knob shall move the telescope socket a minimum of 10 degrees in each direction from the position established in 3.4(e).

3.5.1.3 Limit stops. - The elevation limit stops of the counter shall allow an excursion of 1383 mils in elevation and 50 mils in depression. The limit stop of the horizontal shaft shall allow a minimum movement to 1333 mils in elevation; maximum to 1342 mils.

3.6 Accuracy.

3.6.1 Elevation readings.- With the cross level vial, pitch level vial and elevation level vial bubbles centered, the elevation counter shall indicate quadrant seat elevation settings of zero, 300, 600, 900, and 1200 mils within 0.5 mil excluding backlash.

3.6.2 Horizontal shaft.- The cant level vial bubble shall remain centered, and the line of sight shall not deviate in azimuth by more than 0.25 mil when the horizontal shaft is rotated from 175 mils depression to 1300 mils elevation.

3.6.3 Pitch knob. - With the cant level vial bubble centered, the line of sight shall not deviate from a plumb line by more than 0.5 mil when the pitch knob is actuated throughout its range of movement.

3.6.4 Backlash.

3.6.4.1 Pitch and cant mechanisms.- Backlash in the pitch and cant mechanisms shall not exceed 1.0 mil at zero positions.

3,6.4.2 Elevation reading mechanism.- Backlash in the elevation reading mechanism (counters) shall not exceed 0.5 mil at any elevation or correction setting.

3.7 Rigidity.- With the horizontal shaft immobilized and the mount positioned in accordance with 3.4, the line of sight established subsequent to the gradual application and release of a 20 pound pressure to the right side of the telescope socket (adjacent to the Plate 7660425 and parallel to the horizontal shaft) shall not deviate in excess of the values specified in Table I subsequent to gradually applying and releasing the 20 pound pressure from the opposite direction. This requirement shall also be met with the mount positioned at 800, 1100, and 1333 mils elevation.

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TABLE I

Elevation (In Mils)	Cant (In Mils)	Maximum Deviation (In Mils)
0	0	.25
800	0	.75
1100	0	1.75
1333	0	3.50

3.8 Correction.

3.8.1 Azimuth correction.- The mount shall make left and right azimuth corrections in accordance with the values in Table II within the tolerances specified.

TABLE II

AZIMUTH CORRECTION

Quadrant Seat Elevation (In Mils)	Cant Angle (In Mils)	Azimuth Correction (In Mils)	Tolerance (In Mils)
300	88.9	27.0	.3
600	88.9	59.6	.7
900	88.9	108.8	1.5
1200	88.9	216.8	1.8
300	177.8	54.5	.6
600	177.8	120.3	1.0
1100	177.8	342.4	2.0

3.8.2 Correction counter.- Entering a value into the correction counter, then resetting the elevation counter to zero by rotating the elevation handwheel, followed by recentering the elevation level vial bubble by rotation of the horizontal shaft shall cause the gunner's quadrant seat to elevate or depress within 0.5 mil of the value entered into the correction counter. This requirement shall be met with plus and minus values of 1, 12, 27, and 50 mils entered into the correction counter.

3.9 Operating torques.- The maximum operating torques of the controls shall be within the values specified in Table 111 when checked at the temperatures specified therein. The toggle switch shall be operable at all of the temperature specified in Table III.

TABLE III

Control	OPERATING TORQUE VALUES 60° to 90°F and 150°F	-40°F
Cant correction knob	3 to 6 lb - inches	3 to 12 lb - inches
Elevation handwheel	8 to 12 lb. - inches	15 lb - inches max.

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Elevation correction knob	2 to 4 lb-inches	2 to 8 lb-inches
Quadrant level knob	4 to 10 oz-inches	4 to 32 oz-inches
Pitch knob	5 to 16 lb-inches	5 to 25 lb-inches

3.10 Electrical.

3.10.1 Illumination.- The bulbs for the level vials and the counters, when energized with 18 volts then 28 volts, shall provide sufficient illumination at each voltage for the level vial markings and counter dial readings to be clearly defined when observed in a darkened area.

3.10.2 Insulation resistance.- Insulation resistance shall not be less than 10 megohms with a potential of 100 volts direct current applied.

3.10.3 Continuity.- The continuity of the wiring shall meet the applicable wiring and continuity drawing requirements.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4,1.1 General provisions.- The component and subassembly inspection requirements of MIL-F-13926 form a part of the Quality Assurance Provisions of this specification. Definitions of inspection terms shall be as listed in MIL-STD-109.

4.2 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of three (3) telescope mounts plus three (3) each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.2.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

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4.2.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor's inspection.

4.3 Inspection provisions.

4.3.1 Submission of product.- Unless otherwise specified by the contracting officer, inspection lot size, lot formation and presentation of lots shall be in accordance with "Submission of Product" provisions of MIL-STD-105.

4.3.2 Examination and tests.

4.3.2.1 Components and subassemblies.- All components and subassemblies shall be inspected in accordance with the inspection provisions contained in Supplementary Quality Assurance Provisions (SQAP) provided with the technical data package (TDP). In the absence of SQAP's, the applicable Quality Assurance Provisions of MIL-F-13926 shall apply.

4.3.2.2 Final-acceptance inspection.- Subsequent to first article approval, examination and tests related to Section 3 herein shall be performed on a single defect (individual characteristic) basis in accordance with MIL-STD-105 and the sampling plan specified in Table IV below. Examination and tests for Packaging, Packing and Marking shall be in accordance with MIL-P-14232 and Section 5 herein. The tabulated classification of defects in Table IV and V shall constitute the minimum inspection to be performed by the supplier after first article approval and prior to Government acceptance or rejection by item or lot.

TABLE IV - CLASSIFICATION OF DEFECTS

CLASS: CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
CRITICAL: NONE		
MAJOR: AQL 1.0% Defective		
101. Vibration	3.3.2	4.6.2.1
102. Excursion ranges (cant)	3.5.1.1	4.6.2.3.1
103. Excursion ranges (pitch)	3.5.1.2	4.6.2.3.2
104. Excursion ranges (limit stops)	3.5.1.3	4.6.2.3.3
105. Accuracy elevation readings	3.6.1	4.6.2.4.1
106. Accuracy horizontal shaft	3.6.2	4.6.2.4.2
107. Accuracy of pitch mechanism	3.6.3	4.6.2.4.3
108. Backlash pitch mechanism	3.6.4.1	4.6.2.4.3
109. Backlash cant mechanism	3.6.4.1	4.6.2.7
110. Backlash elevation mechanism	3.6.4.2	4.6.2.4.1
111. Rigidity	3.7	4.6.2.6

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112.	Correction azimuth	3.8.1	4.6.2.8
113.	Correction counter operation	3.8.2	4.6.2.5
114.	Torque cant mechanism (60°-90°F)	3.9 (Table III)	4.6.2.9
115.	Torque elevation mechanism (60°-90°F)	3.9 (Table III)	4.6.2.9
116.	Torque correction mechanism (60°-90°F)	3.9 (Table III)	4.6.2.9
117.	Torque quadrant level mechanism (60°-90°F)	3.9 (Table III)	4.6.2.9
118.	Torque pitch mechanism (60°-90°F)	3.9 (Table III)	4.6.2.9
119.	Illumination	3.10.1	4.6.2.10
120.	Insulation resistance	3.10.2	4.6.2.11
121.	Continuity	3.10.3	4.6.2.12

MINOR: NONE

4.3.2.3 Acceptance and rejection.- Rejected lots shall be screened for all defective characteristics. Removal or correction of defective units and resubmittance of rejected lots shall be in accordance with "Acceptance and Rejection" as specified in MIL-STD-105,

4.3.3 Certification. - Certification shall include test data and results for all characteristics specified below. Certification will be required prior to performing final acceptance inspection and shall suffice for the remainder of the production quantities, provided the manufacturing processes and techniques used to produce the items for which certification was issued have not been changed or revised. Any and all changes will raequire a new certification from the contractor.

CHARACTERISTIC	TEST DATA TO COMPLY WITH
Fabrication	3.1
General specification	3.2

4.4 Special sampling for environmental testing.

4.4.1 Special sampling.- Sampling inspection for characteristics listed in the following table shall be in accordance with inspection level S3, Table I of MIL-STD-105. Acceptance or rejection of the lot shall be in accordance with AQL 2.5 of Table II-A of M1L-STD-105. Samples shall have met all the requirements and tests outlined in Table IV prior to being subjected to the following examination and tests.

TABLE V

CHARACTERISTIC	REQUIREMENT	TEST PROCEDURE
301. Temperature (-40° +150°F)	3.3.1	4.6.1.1
302. Torque cant mechanism (-40° +150°F)	3.9 (Table III)	4.6.2

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303.	Torque elevation mechanism (-40°, +150°F)	3.9 (Table III)	4.6.2.9
304.	Torque correction mechanism (-40°, +150°F)	3.9 (Table III)	4.6.2.9
305.	Torque quadrant level mechanism (-40°, +150°F)	3.9 (Table III)	4.6.2.9
306.	Torque pitch mechanism (-40°, +150°F)	3.9 (Table III)	4.6.2.9

4.4.2 Failure of sample.- Should any one item of a special sampling fail to meet the specified test requirements, acceptance of the represented inspection lot will be suspended by the Government until necessary corrections have been made by the contractor and the resubmitted samples have been approved (see 4.3.2,3).

4.5 Inspection equipment.- Except as otherwise provided for by the contract, the contractor shall supply and maintain inspection equipment in accordance with the applicable requirements of MIL-I-45607.

4.5.1 Government furnished inspection equipment.- Where the contract provides for Government furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements specified in MIL-I-45607.

4.5.2 Contractor furnished inspection equipment.

4.5.2.1 Government design.- Unless otherwise specified in the contract, all inspection equipment specified by drawing number in specifications or SQAP forming a part of the contract shall be supplied by the contractor in accordance with technical data provided with the Technical Data Package (TDP).

4.5.2.2 Contractor design.- The contractor shall design and supply inspection equipment compatible with the "Test Methods and Procedures" specified in 4.6 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926. Since tolerance of test equipment is normally considered to be within 10% of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions and tolerance used in the design of test equipment shall be so selected and controlled as to insure that the test equipment will reliably indicate acceptability of a product which does not exceed 90% of the prescribed tolerance limit, and permit positive rejection when non-conforming. Construction shall be such as to facilitate routine calibration of test equipment.

4.5.3 Test equipment.- In conjunction with 4.5.2.2, the following standard test equipment shall be utilized in the performance of the applicable tests as specified in 4.6.

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NOMENCLATURE	DESCRIPTION
1. Hot and cold chamber	Standard type conforming to the accuracies outlined in test facilities specified under test facilities of MIL-F-13926.
2. Torque wrench and scale	Standard shop type accurate to within 10% of drawing tolerance.

4.6 Test methods and procedures.

4.6.1 Environmental.

4.6.1.1 Temperature test.- The telescope mount shall be subected to Test Procedure I of MIL-F-13926; however, the temperatures specified in 3.3.1 shall apply and the extreme temperatures of -80°F and +160°F shall be excluded from this procedure. When the telescope mount has been stabilized at the temperature specified, the result of the specified test shall conform to the limits outlined in 3.3.1 using test procedures outlined herein.

4.6.2 Performance.

4.6.2.1 Vibration.- The telescope mount shall be vibrated in accordance with Test Procedure V of MIL-F-13926, however, the duration frequency and amplitude shall be as specified in 3.3.2. After completion of the vibration test, the mount shall be inspected for broken and damaged parts using visual and tactile inspection techniques to assure compliance with 3.3.2.

4.6.2.2 Orientation.- To assure proper orientation of the fixture and mount, use set-up instruction on Drawing F8599906, and the targets shown as Figures 1 and 2."

4.6.2.3 Excursion ranges.

4.6.2.3.1 Cant.- With the mount positioned as outlined in 4.6.2.2, rotate the cant knob maximum clockwise direction and position the inspection aid over the proper set of pins of the sighting device. read the cant angle when inspection aid level vial indicates level. This angle must conform to the limits specified in 3.5.1.1. Repeat this procedure after rotating the cant knob to the maximum counter-clockwise position and observe the angular reading for conformance with 3.5.1.1.

4.6.2.3.2 Pitch. - With the mount positioned as outlined in 4.6.2.2, rotate the pitch knob maximum clockwise direction and position the inspection aid over the proper set of pins of the sighting device. Read the pitch angle when the inspection aid indicates level for conformance with 3.5.1.2, Repeat this procedure after rotating the pitch knob to the maximum counter-clockwise position and observe the angular reading for conformance with 3.5.1.2.

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4.6.2.3.3 Limit stops.- With the mount positioned as outlined in 4.6.2.2, rotate the elevation knob of the mount fully clockwise and then fully counterclockwise. Observe both counter readings for conformance with the limits specified in 3.5.1.3. Return the elevation counter to zero mils. Rotate the elevation handwheel of the fixture until the limit stop is contacted. Rotate the quadrant level knob to indicate a cross level position. Seat the inspection aid on the quadrant seat and measure the elevation angle on the inspection aid when the level vial indicates a level position. This measured angle must conform to the limits outlined in 3.5.1.3. Repeat this procedure in the opposite direction.

4.6.2.4 Accuracy and backlash.

4.6.2.4.1 Elevation. - With the mount positioned as specified in 4.6.2.2, set the inspection aid (preset to one of the selected elevation readings specified in 3.6.1) on the quadrant support assembly. By means of the fixture elevation handwheel, induce the selected elevation setting into the mount. True elevation is assured when the inspection aid indicates level. Recheck quadrant level vial for a level reading. Rotate the elevation knob and set the mount's elevation level to indicate level. Record the counter reading. Continue to rotate the elevation fixture handwheel 50 mils beyond the level setting and then reverse the rotation of the elevation handwheel until the level vial of the inspection aid indicates level. Do not override this position. Rotate the elevation knob 50 mils beyond the previous setting. Return the elevation level vial to indicate level. Record this counter reading. Both recorded elevation readings shall coincide with the preset elevation readings of the inspection aid within the limits outlined in 3.6.1. The difference, if any, between the counter reading and the inspection aid must be within the limits specified in 3.6.4.2. Repeat this procedure at the elevation settings specified in 3.6.1,

4,6.2.4.2 Horizontal shaft.- With the fixture and mount positioned as outlined in 4.6.2.2, sight through the auxiliary telescope (5549108) and observe coincidence between the intersecting point of the reticle crosslines and the plumb line while rotating the fixture elevation handwheel through the elevation range specified in 3.6.2. During this motion, the coincided points shall remain in coincidence within the thickness of the reticle line and the cant level vial shall remain level.

4.6.2.4.3 Pitch knob. - With the mount and fixture positioned as outlined in 4.6.2.2, sight through the auxiliary telescope (5549108) and observe coincidence between the intersecting point of the reticle crosslines and the plumb line while rotating the fixture elevation handwheel in one direction and the pitch knob in the opposite direction. They must remain in coincidence within twice the thickness of the reticle line to assure compliance with 3.6.3 when actuated through the excursion specified in 3.6.3. Return the mount and fixture to the position outlined in 4.6.2.2. With the inspection aid set to zero elevation and placed over the proper set of pins, observe coincidence between the elevation level vial and the inspection aid level vial. They must coincide within the limits specified in 3.6.4.1 at zero pitch. Make all measurements on the inspection aid. Scribe an index line on the knob and a mating line on the casting in line with the previously scribed line after zero pitch has been established.

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Record this reading. Rotate the pitch knob one revolution in the clockwise direction and then return the knob to realign the scribe lines. Do not override this point of coincidence. Observe coincidence between the level vial readings. They must coincide within the limits specified in 3.6.4.1. Repeat this procedure by rotating the pitch knob in the counterclockwise direction. The difference in readings, if any, is considered backlash and must be within the limits of 3.6.4.1.

4.6.2.5 Correction counter.- With the mount positioned as outlined in 4.6.2.2, insert one of the specified correction factors into the elevation counter. Inserting a positive correction will be indicated as a decrease in the elevation counter reading. Inserting a negative correction will be indicated as an increase in the elevation counter reading. When one of the specified correction factors has been inserted into the counter by means of the correction counter knob, reset the elevation counter to zero by means of the elevation knob. Actuate the fixture elevation handwheel and relevel the elevation level bubble. Preset the inspection aid to the correction factor over the quadrant support assembly. Observe coincidence between the elevation level vial and the inspection aid level vial. They must coincide within the limits specified in 3.8.2. Make all adjustments on the inspection aid.

4.6.2.6 Rigidity. - With the mount and fixture positioned as outlined in 4.6.2.2, sight through the auxiliary telescope and observe coincidence between the center vertical reticle line, plumb line and target line A of Figure I. They must coincide. Using a push pull type gage against the telescope socket as specified in 3.7, apply a 20 pound push and remove the pressure. Record where the intersecting point of the collimator reticle falls with respect to target line A. Repeat this procedure from the opposite direction and record where the intersecting point of the collimator reticle falls with respect to target line A. Repeat this process using th efixture elevation handwheel to insert the elevation specified in Table I. Any deviation for a selected elevation must conform to 3.7.

4.6.2.7 Cant mechanism backlash.- With the fixture and mount positioned as outlined in 4.6.2.2, remove the securing medium of the sighting device (4 screws) and rotate the telescope's line of sight 90 degrees from the previously established position (4.6.2.2) and resecure the telescope by means of the 4 screws. Recheck level and cross level position of the telescope's line of sight by placing the inspection aid set to zero over the appropriate sets of pads on the sighting device. Erect a plumb line approximately 40 feet from the telescope's objective lens and in line with the intersecting point of the telescope's center reticle lines. Rotate the telescope in the support and align the center vertical reticle line in coincidence with the plumb line. Position target Figure 2 approximately fifty feet from the telescope's objective lens and align target line A to the previously coincided vertical reticle line and plumb line. Raise or lower target and coincide target line B to the horizontal reticle line. Rotate the cant knob through the maximum excursion and rotate the fixture cross leveling handwheel in the opposite direction while sighting through

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the auxiliary telescope and observe coincidence between target line A and the intersecting point of the telescope center horizontal and vertical reticle lines. They must remain in coincidence within the thickness of the reticle line. Return the mount to zero position and relevel all level vials. Observe coincidence between the cant indicator (red line) and housing index line. Consider width of both lines when determining coincidence. Rotate the cant knob one revolution and reverse rotational direction until lines are in coincidence (do not override). Sight through the telescope and observe where the intersecting point of the telescope's reticle lines fall upon the target with respect to tolerance zone BC or DB (depending on the direction of rotation). Record this position. Continue to rotate the cant knob beyond this point one revolution and return the knob and realign the previously coincided index lines (do not override). Observe and record where the intersecting point of the telescope's reticle lines fall with respect to tolerance zone BC or BD (depending on the direction of rotation). The difference between the recorded positions must comply with limit specified in 3.6.4.1.

4.6.2.8 Azimuth correction.- With the mount and fixture positioned as in 4.6.2.7, preset the inspection aid to first cant reading specified in Table II and position the inspection aid over the seating surfaces of the sighting device. By means of the fixture cross leveling handwheel, set the inspection aid level vial to a level position (do not override). By means of the mount cant knob, relevel the mount's cross level bubble. Preset the first elevation setting specified in Table II into the inspection aid and position the inspection aid on the seating surfaces of the quadrant support assembly. By means of the fixture elevation handwheel, reposition the inspection aid level vial to indicate level (do not override). Relevel the quadrant support level vial by the support knob and if necessary relevel the inspection aid level vial by means of the fixture elevation handwheel. Remove the locating pin from the zero position of the sighting device and insert this pin into the appropriate locating hole (27.0). Observe coincidence between the intersecting point of the telescope center reticle line and taget line A. They must coincide within the limits of tolerance line "a". Repeat this procedure at each of the settings specified in 3.8.1. The telescope's "intersecting point must always coincide with target line A within the limits of tolerance lines for a selected combination of elevation and cant angles specified in Table II. The sighting device must also be changed for-each combination of elevation and cant angles by inserting the locating pin into the appropriate set of holes.

4.6.2.9 Operating torque. - Torque testing shall be performed using appropriate torque measuring equipment and adapters and shall be performed at the temperature specified in Table III. The measured torque shall conform to the values specified in 3.9.

4.6.2.10 Illumination.- With a power supply of the voltage specified in 3.10.1 applied to the mount, observe brightness of the level vial and counters in a darkened area for conformance with 3.10.1.

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4.6.2.11 Insulation resistance.- This test to be performed as outlined in Test Method 302 of MIL-STD-202 using test condition A. The test shall be made between insulated points and ground. Resistance shall be as specified in 3.10.2.

4.6.2.12 Continuity.- Use standard shop type continuity checker in conjunction with standard inspection techniques to determine conformance with 3.10.3.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. - Packaging, packing and marking shall be in accordance with Specification MIL-P-14232 and Packaging Data Sheet 8267701. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The Telescope Mount M145 is primarily intended to support the Panoramic Telescope M117 in the 105mm Howitzer M108 and the 155mm Howitzer M109. The mount provides cant correction and elevation movement for the panoramic telescope. The parallelogram linkage between the telescope mount and the gun trunnion is not part of the mount but a component of the particular fire control system that adapts the mount to the model vehicle in which it is utilized.

6.2 Order data.- Procurement documents should specify the following:

(a) Title, number an date of this specification.
(b) Selection of an applicable level or preservation, packaging, and packing.
(c) Applicable stock number.
(d) Applicable packaging data sheet number (see 5.1).
(e) Requirement for first article inspection.

Custodian:

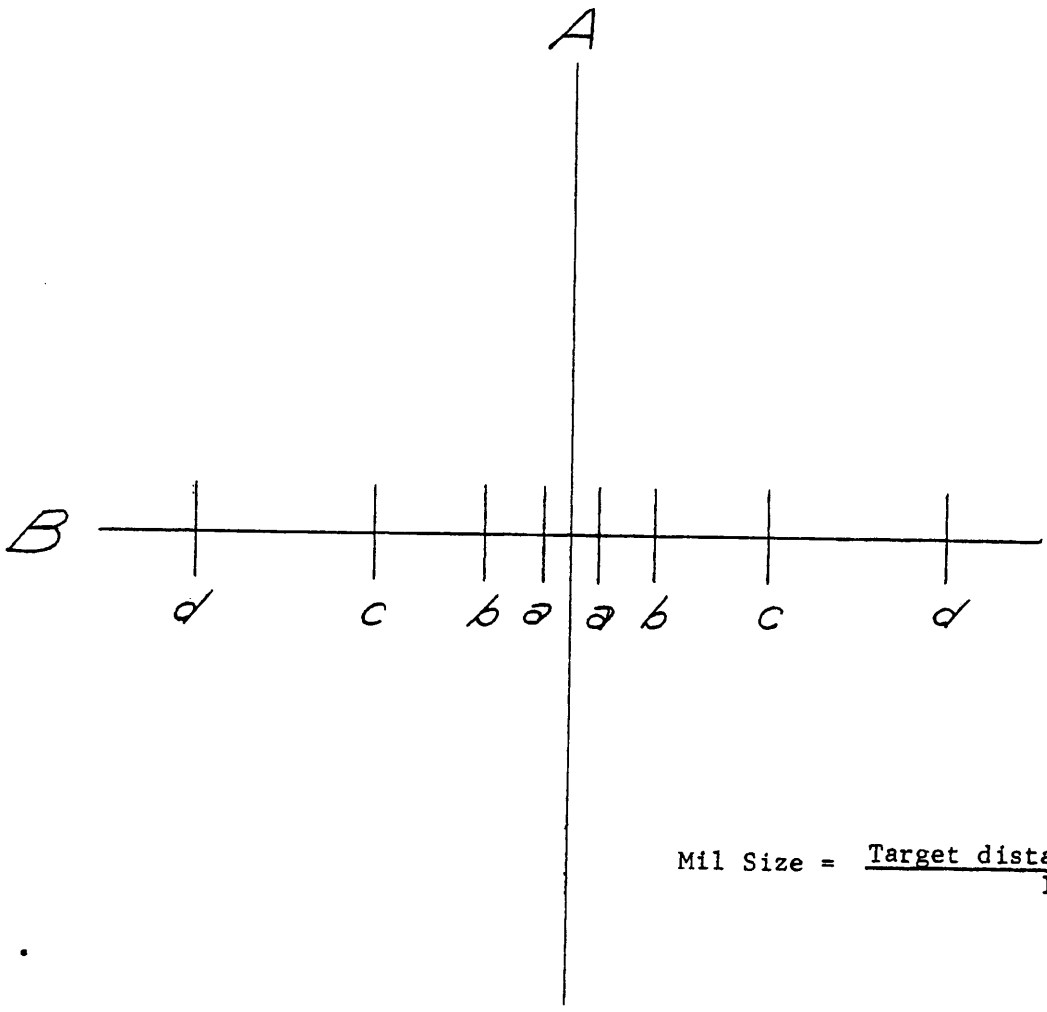
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Preparing activity:

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$$\text{Mil Size} = \frac{\text{Target distance (In inches)}}{1000}$$

Spaces for 50' TD
a-A = .15"
b-A = .45"
c-A = 1.05"
d-A = 2.1"

FIGURE I

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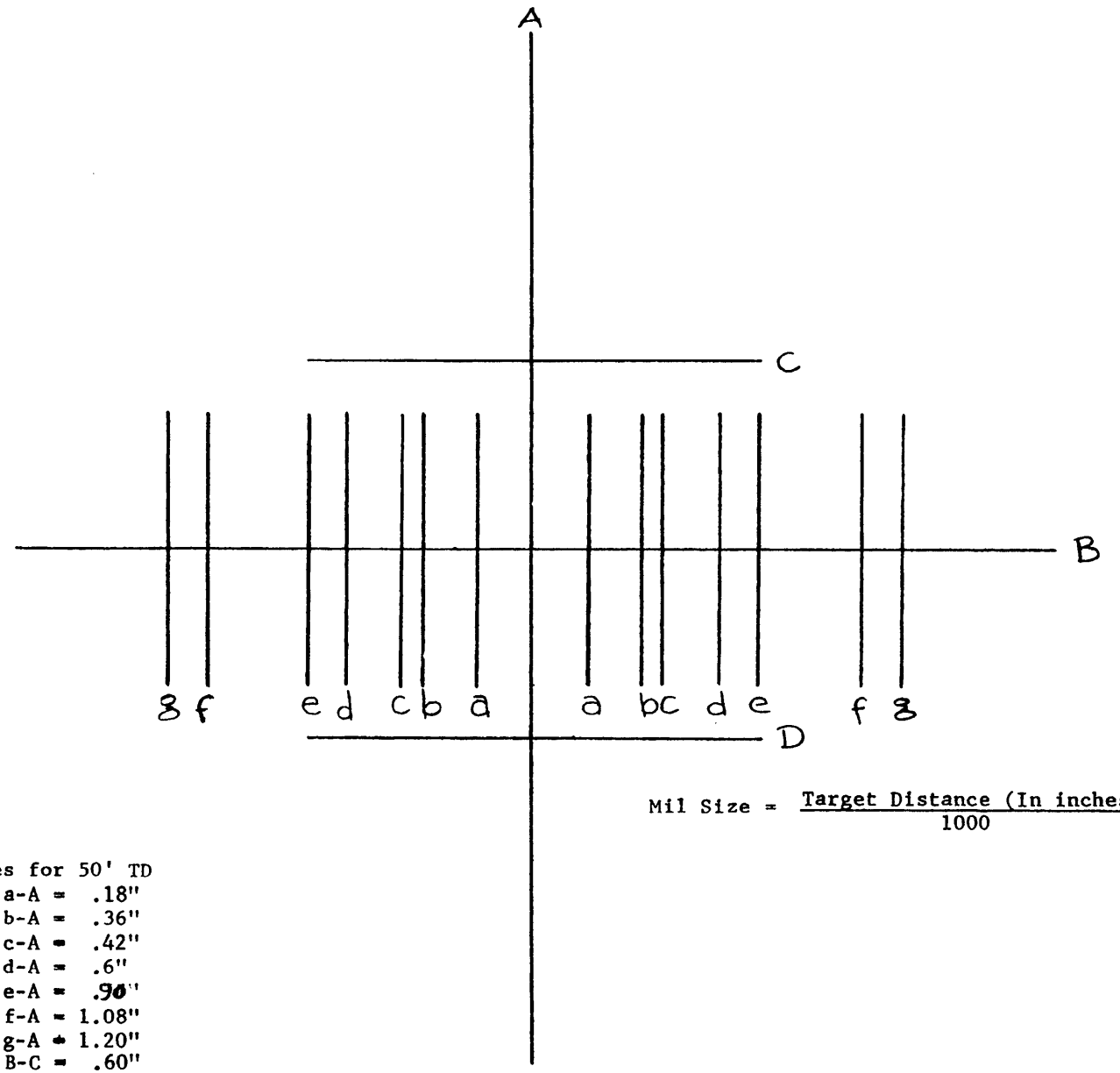


FIGURE 2.

Mil Size = $\frac{\text{Target Distance (In inches) (TD)}}{1000}$

Spaces for 50' TD
a-A = .18"
b-A = .36"
c-A = .42"
d-A = .6"
e-A = .90"
f-A = 1.08"
g-A = 1.20"
B-C = .60"
B-D = .60"

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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